DEEP LEARNING

REPORTE PROYECTO 2

GANs

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Introduction

Generative Adversarial Networks (GANs) are a class of deep learning models introduced by Goodfellow et al. (2014) that learn to generate realistic synthetic data by training two neural networks in opposition: a **Generator** that produces fake samples from random noise, and a **Discriminator** that tries to distinguish real samples from generated ones. Through this adversarial process, the generator progressively improves until its outputs are indistinguishable from real data.

This project explores GANs in two stages, following a natural progression from unconditional to conditional generation:

- A Simple GAN based on the DCGAN architecture (Radford et al., 2015), which generates face images from random noise without any control over the output content.
- AttGAN (He et al., 2019), an attribute-conditional GAN that allows specific facial features — such as hair color, the presence of glasses, or apparent age — to be modified in an existing image while preserving the person's identity.

The goal is not merely to generate realistic faces, but to understand how conditioning information can give a model precise editorial control over the output, a capability with broad applications in image editing, data augmentation, and creative tools.

Objectives

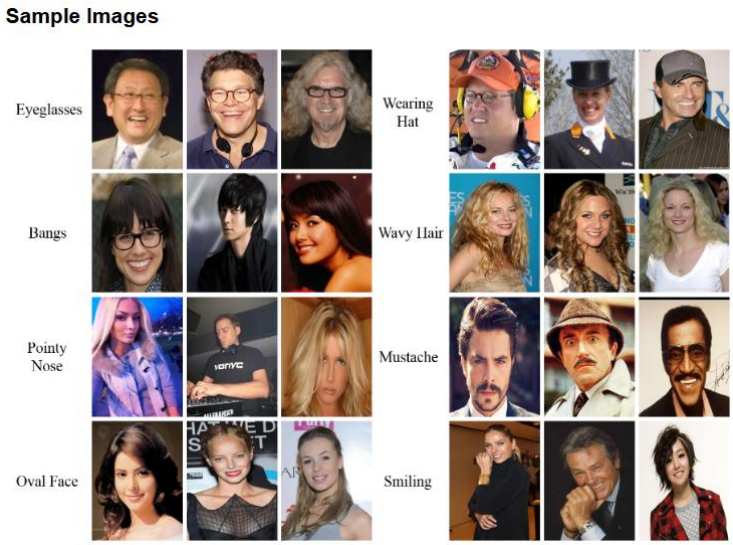
1. Understand the fundamental principles of Generative Adversarial Networks and their role in generative deep learning, including the adversarial training dynamic and common failure modes.
2. Implement a baseline unconditional GAN (DCGAN) on the CelebA dataset and observe how image quality evolves over training epochs.
3. Implement AttGAN, a conditional GAN variant designed for facial attribute editing, and compare its behaviour and output quality to the simple baseline.
4. Design and run three controlled experiments with AttGAN by varying the loss weight hyperparameters and analyse how each configuration affects the trade-off between identity preservation and attribute edit strength.
5. Evaluate all models quantitatively using FID (Fréchet Inception Distance) and DACID (our custom metric), and qualitatively through visual inspection of generated and edited images.

Dataset

The CelebFaces Attributes Dataset (CelebA) is a large-scale benchmark for face-related tasks, widely used in the GAN literature. It was introduced by Liu et al. (2015) and contains over 200,000 celebrity face images with rich annotations.

CelebA at a glance	Key statistics
Total images	202,599 face photographs
Unique identities	10,177 celebrities
Attribute labels	40 binary attributes per image
Landmark points	5 facial keypoints per image
Official splits	Train 162,770 · Valid 19,867 · Test 19,962

The dataset covers large variations in pose, lighting, expression, and background, making it a challenging and realistic benchmark. Its 40 binary attribute annotations make it especially suitable for conditional generation and attribute editing tasks.



Attribute Selection

For this project we selected 13 attributes from the 40 available, chosen to cover a representative and diverse range of facial characteristics:

Category	Attributes
Hair	Bald, Bangs, Black_Hair, Blond_Hair, Brown_Hair
Facial hair	Mustache, No_Beard, Bushy_Eyebrows
Accessories	Eyeglasses
Demographics	Male, Young, Pale_Skin
Expression	Mouth_Slightly_Open

Preprocessing

All images are preprocessed uniformly before being fed to either model:

- Center-cropped to 178×178 pixels to remove excess background above the forehead and below the chin.
- Resized to 128×128 pixels (AttGAN) or 64×64 pixels (Simple GAN via bilinear interpolation during training).
- Randomly horizontally flipped for data augmentation during training.
- Normalised to the range [−1, 1] using mean=0.5 and std=0.5 per channel, matching the Tanh output activation of both generators.
- Attribute labels are converted from binary {0, 1} to bipolar {−1, +1} for use as conditioning vectors.

Methodology

Simple GAN (DCGAN)

Before introducing the complexity of conditional generation, we first implemented an unconditional DCGAN as a baseline. This serves two purposes: it validates that our training pipeline and data loading are correct, and it provides a reference point for comparing image quality and training stability against AttGAN.

Architecture

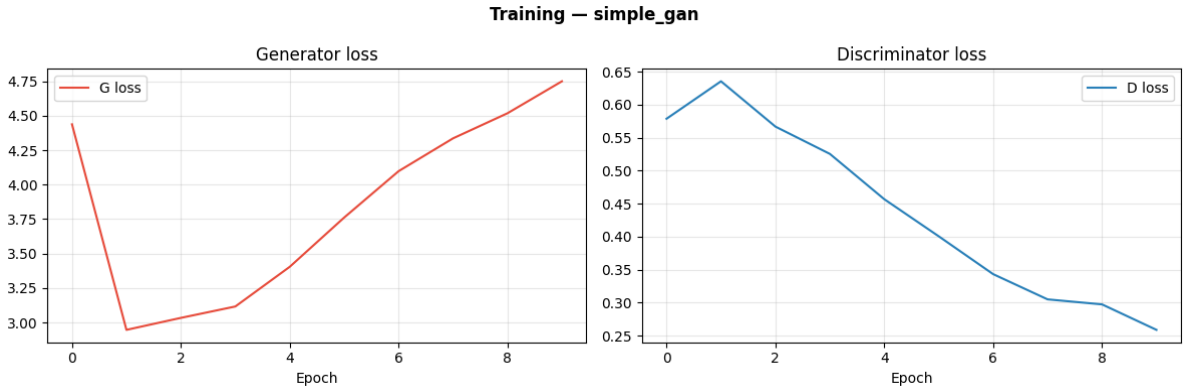
The Simple GAN follows the DCGAN guidelines of Radford et al. (2015), using strided convolutions, batch normalization, and specific activation functions at each stage:

Component	Input	Output	Key layers
Generator	z (100×1×1)	image (3×64×64)	5× ConvTranspose2d + BatchNorm + ReLU → Tanh
Discriminator	image (3×64×64)	scalar [0,1]	5× Conv2d + BatchNorm + LeakyReLU(0.2) → Sigmoid

Loss Function

The Simple GAN uses standard binary cross-entropy adversarial loss.

Results




```
[metrics] Extracting features (512 images each)...
[metrics] FID   = 64.6355
[metrics] DACID = 4.1346
FID   : 64.6355
DACID : 4.1346
```

AttGAN

The Simple GAN can generate faces from noise but offers no control over the output. AttGAN (He et al., 2019) addresses this by conditioning the generator on a target attribute vector, allowing precise editing of specific facial attributes while preserving the overall identity and non-targeted regions of the face.

Architecture

AttGAN uses an encoder–decoder architecture with a multi-task discriminator:

Loss Functions

AttGAN is trained with three losses working together:

Training Procedure

Training alternates between a discriminator step and a generator step for each batch:

1. Sample a batch of real images and their attributes.
2. Shuffle the attribute batch to create target attributes a' (a different identity's attribute profile).
3. Train D: forward pass real images (adv + cls loss), generate fakes with no gradient, forward fakes through D (adv loss only).
4. Train G (Encoder + Generator): compute reconstruction with original attributes, compute translation with target attributes, update using combined loss.

References

I. Goodfellow et al., "Generative Adversarial Nets," NeurIPS 2014. https://arxiv.org/abs/1406.2661